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2007 J. Phys. B: At. Mol. Opt. Phys. 40 2809

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Light grating storage in cold atoms

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Received 9 March 2007, in final form 17 May 2007

Published 4 July 2007

Online at stacks.iop.org/JPhysB/40/2809

Abstract

We report the storage of a spatial light grating in a sample of cold caesium atoms. The grating is stored in the atomic medium using a backward four-wave-mixing laser beam configuration with two counterpropagating control fields and a noncollinear probe field. The grating was stored during a controllable dark interval after which turning on the control fields alone resulted in a light pulse emission in the forward and backward probe beam directions. Grating storage times up to 4 μ s were observed.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

The light storage (LS) effect, allowing the retrieval of a light pulse from an atomic medium depending on the previous incidence of a probe pulse in the medium, has attracted very much interest in the past few years [1, 2]. In most experiments, LS is observed under conditions of electromagnetically induced transparency (EIT) [3, 4]. The medium is made transparent to the propagation of the probe pulse due to the presence of a pump field. In the LS sequence, the pump field is used to control the propagation, storage and retrieval of the probe pulse. Initially, the control and the probe pulses propagate simultaneously in the atomic medium, afterwards the control field is turned off resulting in the storage of information carried by the probe field in the quantum coherence between atomic states of the sample. Switching back on the control field results in the release of a light pulse sharing many common features with the probe pulse. The simplest system allowing the observation of EIT and LS is a sample of three level atoms with two ground (or metastable) levels and an excited state. The two transitions from the ground levels to the excited state are coupled respectively to the pump and probe fields (Λ system). In this system, the information concerning the incident fields is stored in the atomic coherence between the two lower states. The storage time is limited by the lower states dephasing rate. EIT and LS have been observed using as lower levels two states belonging to different hyperfine levels [5] or two Zeeman sublevels [6] of the same hyperfine atomic level.

EIT can be understood as a result of the interaction of the probe field with the coherently prepared atomic sample. If noncollinear pump and probe beams are used, a spatial modulation (grating) of the atomic coherence is obtained. This coherence grating can interact with the incoming fields producing diffracted beams in the phase matching directions. Such electromagnetically induced gratings (EIG) were observed in [7, 8]; transfer of coherence gratings between different hyperfine levels was reported in [9].

To date, LS experiments were performed using collinear pump and probe fields. In most of these experiments, the writing and retrieved fields propagate in the same direction [5, 6, 10, 11]. The possibility of halting and reversing the probe pulse was demonstrated in [12, 13]. In this work, we present an experimental observation of LS using noncollinear pumps and probe fields. This configuration allows us to demonstrate the storage of the light grating produced by the incident fields in the form of a spatial modulation of the atomic coherence. Such coherence grating preserves information on the relative phase and polarizations of the writing fields. Our observation constitutes a first step towards the storage of more complex spatial phase structures, as that associated with beams carrying orbital angular momentum [14, 15].

2. Experimental setup and results

We have investigated the LS process employing the well-known backward four-wave-mixing (FWM) configuration with counterpropagating pump beams and a nearly collinear probe beam. The experiment was performed using a degenerate two-level system corresponding to the cycling transition $6S_{1/2}(F = 3) \leftrightarrow 6P_{3/2}(F' = 2)$ of the caesium D_2 line. The caesium atoms were previously cooled in a MOT operating on the closed transition $6S_{1/2}(F = 4) \leftrightarrow 6P_{3/2}(F' = 5)$ with a repumping beam resonant with the open transition $6S_{1/2}(F = 3) \leftrightarrow 6P_{3/2}(F' = 3)$, as shown in the simplified level scheme in figure 1(a). The temperature of the MOT was estimated to be approximately 1 mK. The trapping beams, provided by a Ti:sapphire laser, have a total power of approximately 80 mW, a diameter of 2.0 cm and are red-detuned 12 MHz from the trapping transition. In order to pump the atoms into the state $6S_{1/2}(F = 3)$, we used a mechanical chopper, with a duty cycle of 95 %, to switch off the repumping beam during a 1 ms interval (see figure 1(c)). At the beginning of this interval, the atoms are optically pumped into the $6S_{1/2}(F = 3)$ level by the trapping beams (via nonresonant excitation to the excited state $F' = 4$). The optical pumping time, measured by monitoring the absorption of a weak probe beam nearly resonant with the $F = 3 \rightarrow F' = 2$ transition, is of the order of 50 μ s. After this time, an optical density up to approximately 3 for the sample of cold atoms on the $6S_{1/2}(F = 3) \leftrightarrow 6P_{3/2}(F' = 2)$ transition can be obtained depending on the MOT parameters.

For the observation of both the EIT and EIG spectra, we use the well-known phase conjugation beam geometry illustrated in figure 1(b). The forward and backward pump beams have opposite circular polarization. The probe beam polarization is circular with opposite handedness with respect to the nearly copropagating forward pump beam. For such a polarization choice, two Zeeman sublevels belonging to the $F = 3$ ground state are coupled to the same Zeeman sublevel of the $F' = 2$ excited state by the incident fields (see the level scheme in figure 1(a)). All the beams are provided by the same external cavity diode laser which is locked to the centre of the $(F = 3) \leftrightarrow (F' = 2)$ transition. Before reaching the MOT, the probe beam passes through a pair of acousto-optic modulators (AOM). One of them, used in double passage, is driven by a variable frequency RF generator and allows the tuning of the probe beam frequency offset with respect to the pumping beams. The pump beam also passes through another pair of AOMs which keep its frequency unchanged, but allows

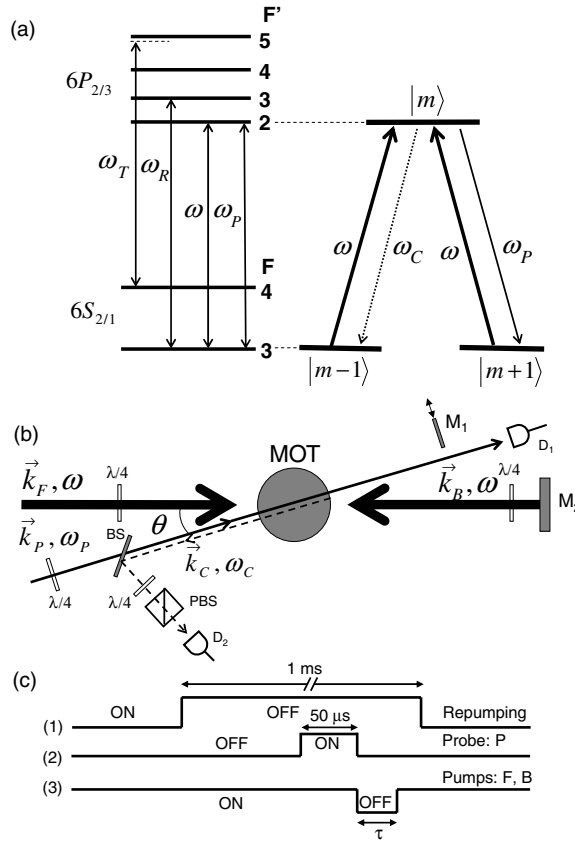


Figure 1. (a) Left: the energy levels of caesium D₂ line. ω_T , ω_R , ω and ω_P are, respectively, the frequency of the trapping, repumping, pumping and probe beams. Right: the relevant Zeeman sublevels coupled by the pumping and probe beams. The arrows represent the parametric four-wave-mixing process in which one photon of each pump beam with frequency ω is absorbed and a pair of photons with frequency $\omega_P = \omega - \delta$ and $\omega_C = \omega + \delta$ is emitted into the probe and conjugate beams, respectively. The atoms are pumped into the lower ground state by switching off the repumping beam accordingly to the time sequence shown in trace (1) of (c). (b) Simplified experimental configuration. The counterpropagating (forward and backward) pumping beams with wave vectors $\vec{k}_F$ and $\vec{k}_B$ ($\vec{k}_F = -\vec{k}_B$) and frequency ω and the probe beam with wave vector $\vec{k}_P$ and frequency ω_P intercept the cold atomic cloud at an angle θ to generate the conjugate beam with wave vector $\vec{k}_C$ and frequency ω_C .

a fast switching on and off. During the acquisition of the spectra, all beam intensities are kept constant in time. The laser beams intercept the cloud of caesium atoms making a small angle $\theta \approx 60$ mrad (figure 1(b)). The backward pump beam is obtained by retro-reflecting the forward beam after traversing a second quarter waveplate, as illustrated in figure 1(b). The pump and probe beam power is 80 μ W and 20 μ W, respectively. The diameter of the pump beams is made much larger than the trapped atom cloud size (≈ 2 mm), while the probe beam is focused inside the trap with a much smaller diameter. Curve (a) in figure 2 shows the probe transmission spectrum (EIT spectrum), as a function of the pump-probe frequency detuning δ , around the $(F = 3) \leftrightarrow (F' = 2)$ transition. The probe scanning ramp is triggered by the turning off of the repumping beam. The same triggering signal is also used to switch off the MOT quadrupole magnetic field. Three orthogonal pairs of Helmholtz coils are used to reduce

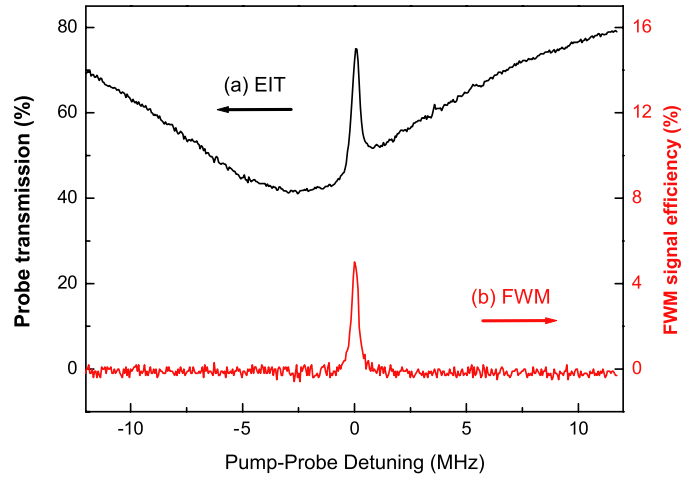


Figure 2. (a) Probe transmission spectrum showing the EIT peak. (b) FWM signal spectrum obtained in the backward probe direction. Both traces were recorded during the 1 ms time interval shown in trace (1) of figure 1(c).

the residual ambient magnetic field to less than 500 mG. The probe transmission is recorded using the photodiode D_1 , while the photodiode D_2 detects the light emitted in the backward probe direction. This is the field diffracted by the EIG. Under stationary excitation such a field corresponds to the probe phase conjugated field due to four-wave mixing of the incident fields. For shortness we refer to the signal recorded by the photodiode D_2 as the FWM signal. The EIT and FWM spectra as a function of the pump-to-probe field detuning are shown in figures 2(a) and (b), respectively. To study the polarization of the probe beam and the reflectivity of the FWM process we use the movable mirror M_1 . Using the polarization analyser placed after the 50–50 BS, we have verified that the polarization of the generated FWM conjugate beam is also circular with opposite handedness with respect to the incident probe beam. This is consistent with the parametric process represented in figure 1(a). The reflectivity of the FWM signal relative to the probe beam intensity is of the order of 5%.

We turn now to the light grating storage experiment. For these observations, the frequency of the probe beam is fixed around the frequency of the pump beam, i.e. ($\delta = 0$), and we measure the transient probe beam transmission and the FWM signal for the time sequence shown in figure 1(c). Within the 1 ms time window for which the repumping beam and the MOT quadrupole magnetic field are turned off, we record the probe transmission and the generated FWM signal during a $50 \mu\text{s}$ interval triggered by the probe pulse. Initially, when both the pump and probe fields are on, a coherence grating induced by the orthogonally polarized forward pump and probe beams is written into the Zeeman ground states of the atomic ensemble. The spatial grating period is given by $\Lambda = \frac{2\pi}{|k_F - k_P|} = \frac{\lambda}{2 \sin(\theta/2)}$. The conjugate signal is then originated from the real time Bragg diffraction of the backward beam into this coherence grating. It is worth noting that the probe and backward beams also form a grating that satisfies the Bragg condition for the diffraction of the forward beam into the probe phase conjugate direction. Since the backward and probe beams have the same circular polarization they induce in this case a population grating. Such a grating has a small spatial period of the order of the optical wavelength and it is expected to be significantly washed out by the atomic motion [8].

At the end of the probe pulse, all beams are turned off and after a controllable dark interval τ the pump beams alone are turned back on. We then observe retrieved probe and FWM pulses

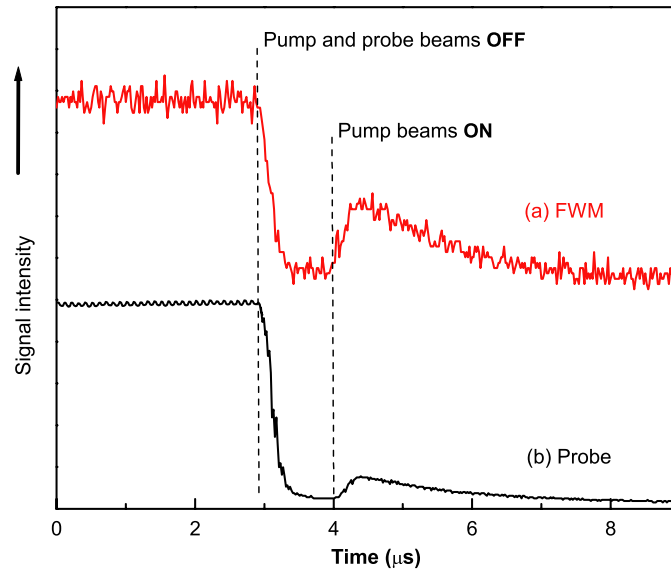


Figure 3. (a) Transient four-wave-mixing signal. The incident forward and backward pump beams and the probe beam are intensity modulated according to the time sequence depicted in traces (2) and (3) of figure 1(c). (b) The corresponding transient probe transmission signal. The conjugate and the probe beams can be retrieved after a controllable time interval.

as shown in figure 3. The retrieval of the probe and FWM beams is due to the Bragg diffraction of the forward and backward pumps, respectively, into the stored coherence grating.

The amplitude of the retrieved pulses relative to the signal level during the writing interval is rather different for the two traces shown in figure 3. It is of the order of 50% for the FWM signal but only a few per cent for the probe transmission. A series of traces of the FWM signal is shown in figure 4 for different dark time intervals. The retrieved FWM was clearly visible after dark time intervals up to 4 μs . The maximum storage time is strongly dependent on the compensation of the residual magnetic field. The observation of retrieved signal pulses in the forward and backward probe directions is a clear indication that a coherence spatial grating was stored in the atomic Zeeman sublevels of the caesium $F = 3$ hyperfine ground state. As in the case of the stationary four-wave-mixing process mentioned above, we have verified that the storage process preserves the polarization of the probe and conjugate beams.

It is worth mentioning that counterpropagating writing and reading pumping beams were also used in the experiments described in [12, 13]. However, in these experiments the signal and control beams are copropagating and, consequently, no spatial ground state coherence grating (with a finite period) is induced or stored in the medium. In consequence, the retrieval process is less sensitive to the atomic motion allowing its observation in hot vapours. In our case, the spatial grating stored in the medium has a period of the order of 14 μm which is easily washed out by the atomic motion in hot vapours. To prevent this washout, the use of a cold-atom sample is essential.

Figure 5 shows the dependence of the fractional intensity of the retrieved FWM and probe signals on the storage time τ . In both cases, the fractional intensity decreases exponentially with similar time constants ($\sim 2.7 \mu\text{s}$). The rise time of the retrieved pulses is determined by

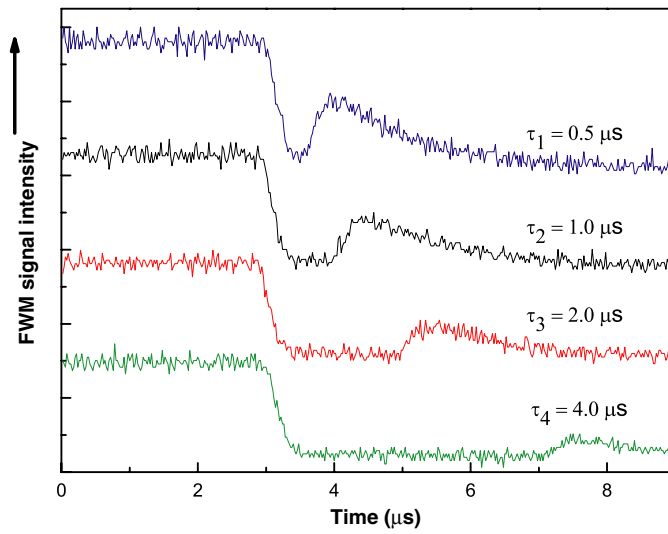


Figure 4. FWM signal for different storage times.

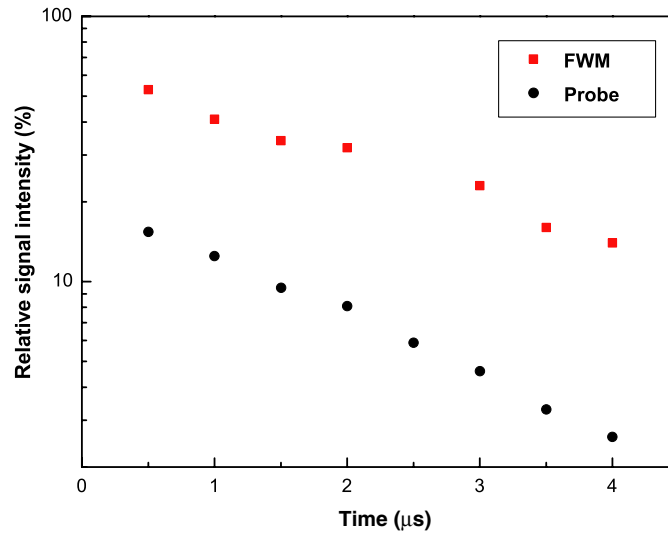


Figure 5. Fractional intensity of the retrieved FWM and probe transmission signals as a function of the dark time interval.

the switching time of the light fields ($0.5 \mu\text{s}$). The decay of the retrieved pulse is exponential with a characteristic time depending on the pump beam intensities [16].

3. Conclusions

In summary, we have investigated the light storage process in an ensemble of cold caesium atoms using a noncollinear four-wave-mixing configuration. We have observed the storage of a polarization grating written into the atomic system as a spatially dependent Zeeman coherence. This observation demonstrates the ability of the atomic system to locally store

information about the relative phase of the incident fields. To our knowledge, this is the first demonstration of light storage within a three-dimensional atomic coherence distribution. This result opens the way to the storage and optical manipulation of complex spatial light field structures such as holographic interference patterns or light modes with unusual symmetries (for instance, modes carrying orbital angular momentum).

Acknowledgments

We gratefully acknowledge Marcos Aurelio for his technical assistance during the experiment and G L Gattobigio for stimulating discussions. This work was supported by the Brazilian Agencies CNPq/PRONEX, CNPq/Inst. Milênio and FINEP.

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